

SOFTWARE ENGINEERING LAB ASSIGNMENT

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Project Title

Food Waste Reduction and Donation Management System

Sustainable Development Goals (SDGs) Covered

- **SDG 2 – Zero Hunger**
Helps provide food to people in need by redistributing surplus food.
- **SDG 12 – Responsible Consumption and Production**
Reduces food wastage by promoting food reuse and donation.
- **SDG 13 – Climate Action**
Reduces environmental pollution and greenhouse gas emissions caused by food waste.
- **SDG 11 – Sustainable Cities and Communities**
Encourages community participation and social responsibility.

Problem Statement

In today's world, a large amount of fresh and edible food is wasted every day by restaurants, hotels, supermarkets, and households. At the same time, many people struggle with hunger and food insecurity.

The main reasons for this problem are:

- Lack of a common platform to connect food donors and NGOs
- Food getting spoiled due to poor planning and late collection
- No proper coordination between donors, volunteers, and NGOs
- Limited awareness about the environmental impact of food waste

As a result, food waste increases, hunger remains unresolved, and environmental pollution grows.

Objectives of the System

- To reduce food wastage at the source
- To provide surplus food to needy people
- To ensure timely collection and safe delivery of food
- To improve coordination between donors, NGOs, and volunteers
- To promote awareness about food waste and sustainability
- To support multiple Sustainable Development Goals

Food Waste Reduction Cycle



Proposed System

The Food Waste Reduction and Donation Management System is a web-based or mobile-based application designed to connect food donors, NGOs, and volunteers on a single digital platform.

How the System Works

1. Food Donors

Restaurants, hotels, supermarkets, and households can register on the system.

When they have extra food, they can list:

- Type of food
- Quantity
- Preparation time

- Expiry time
- 2. Food Listing and Expiry Tracking

The system automatically tracks the expiry time of the listed food.
If food is about to expire, the system gives priority alerts to NGOs and volunteers.
- 3. NGOs and Food Banks

Registered NGOs can view available food donations nearby.
They can:

 - Accept food requests
 - Schedule pickup
 - Manage food distribution to needy people
- 4. Volunteer Coordination

Volunteers can register and provide their availability.
The system assigns volunteers to:

 - Pick up food from donors
 - Deliver food safely to NGOs or shelters
- 5. Pickup and Delivery Management

The system helps in scheduling pickups and tracking delivery status.
Notifications are sent at each step to avoid delays.
- 6. Food Safety and Quality Checks

Basic food safety guidelines are shown to donors and volunteers.
Only fresh and safe food is allowed to be donated.
- 7. Notifications and Alerts
 - New food donation alerts
 - Expiry warnings
 - Pickup reminders
 - Emergency food requests
- 8. Impact Tracking and Reports

The system records:

 - Amount of food saved
 - Number of people helped
 - Reduction in food waste
 - Environmental benefits

Expected Benefits

- Significant reduction in food waste
- Better food distribution to the needy
- Improved community involvement
- Reduced environmental pollution
- Increased social responsibility

